A separable subcase of the cardinality-injective problem

Partial packet for arXiv:1301.3656

Source and target

The source is Lingxin Bao, Lixin Cheng, Qingjin Cheng, and Duanxu Dai, *On universal left-stability of ϵ-isometries*, arXiv:1301.3656 [1]. In Definition 3.1 the source calls a Banach space X *cardinality injective* if there is $\lambda < \infty$ such that, whenever X isometrically embeds into a Banach space Y with $\mathrm{card}(Y) = \mathrm{card}(X)$, the image of X is the range of a projection of norm at most λ. The source proves that this is equivalent to universal left-stability for standard ε-isometries and that dual cardinality-injective spaces are injective. Remark 3.6 then leaves the following general question.

> injective. But it is the same proof of the necessity of the theorem above. □
>
> **Remark 3.6.** Cheng, Dai, Dong and Zhou [2] recently showed that every injective Banach space is universally left-stable. By this result and Corollary 2.8, we get that a dual Banach space is injective if and only if it is universally left-stable. This and Theorem 3.3 entail that a dual Banach space is cardinality injective if and only if it is injective. But we do not know whether it is true in general.

Thus the target is whether every cardinality-injective Banach space is injective, in the usual bounded-extension/P_λ sense used by the source.

Result

Theorem 1. *Let X be a separable Banach space. If X is cardinality injective in the sense of Bao–Cheng–Cheng–Dai, then X is finite-dimensional. Hence, for separable Banach spaces, the source question has a positive answer: cardinality injective implies injective.*

Proof intuition

For an infinite-dimensional separable space X, the cardinality restriction in Definition 3.1 is no restriction on separable superspaces: every infinite-dimensional separable Banach space has cardinality continuum. Therefore cardinality injectivity makes X separably injective. Zippin's classification then says that X must be isomorphic to c_0. But c_0 cannot satisfy Definition 3.1, because it is an isometric subspace of ℓ_∞, $\mathrm{card}(c_0) = \mathrm{card}(\ell_\infty)$, and Phillips' theorem says there is no bounded projection $\ell_\infty \to c_0$. This rules out the infinite separable case. Finite-dimensional spaces are injective by the standard Hahn–Banach coordinate-extension argument.

Proof

Assume first that X is infinite-dimensional and separable. Then $\mathrm{card}(X) = 2^{\aleph_0}$. If Y is any infinite-dimensional separable Banach space containing X, then likewise $\mathrm{card}(Y) = 2^{\aleph_0}$. Hence the defining projection property for cardinality injectivity applies to every separable superspace Y of

X. Equivalently, X is separably injective: every bounded operator from a closed subspace of a separable Banach space into X extends to the whole separable space with a uniform bound.

By Zippin's theorem [3], a separable separably injective Banach space is isomorphic to c_0. Proposition 3.2 of [1] says that cardinality injectivity is preserved under Banach-space isomorphism. Therefore, if our infinite-dimensional separable X were cardinality injective, then c_0 would be cardinality injective.

This is impossible. The standard inclusion $c_0 \subset \ell_\infty$ is isometric, and both spaces have cardinality $2^{\aleph_0}$. Cardinality injectivity of c_0 would therefore give a bounded projection $\ell_\infty \to c_0$. Phillips' theorem [2] says that no such projection exists. This contradiction proves that no infinite-dimensional separable cardinality-injective Banach space exists.

It remains only to note that finite-dimensional Banach spaces are injective in the bounded-extension sense used in the source. Let X be n-dimensional and isometrically contained in a Banach space Y. Choose an Auerbach basis $(x_i, x_i^*)_{i=1}^n$ for X. Extend each x_i^* to $\phi_i \in Y^*$ by Hahn–Banach with $\|\phi_i\| = \|x_i^*\| = 1$, and set

$$P(y) = \sum_{i=1}^{n} \phi_i(y) x_i \qquad (y \in Y).$$

Then $P : Y \to X$ is a projection and $\|P\| \le n$. Thus finite-dimensional spaces are P_n-spaces, hence injective in the sense of the source. The separable case follows.

Verification and limitations

The local run indexes were searched for the exact arXiv id, for `cardinality injective`, and for universal left-stability keywords. No existing packet for arXiv:1301.3656 was found. A neighboring attempt for arXiv:1301.3396 concerns broader epsilon-isometry stability questions and not Remark 3.6. The local later source arXiv:2605.06003 mentions the same cardinality-injective condition but does not settle this question. Web searches for exact phrases around cardinality injectivity found the source paper and no later exact full resolution during this pass.

This packet proves only the separable subcase. It does not settle the nonseparable cardinality-injective versus injective problem.

Human-review recommendation

Likely valid partial result. Review should check that the source's "injective" convention is the bounded P_λ/extension convention, that Proposition 3.2 is being used with the same isomorphic notion, and that the cited Zippin and Phillips theorems are acceptable standard inputs.

References

[1] L. Bao, L. Cheng, Q. Cheng, and D. Dai, *On universal left-stability of ϵ-isometries*, arXiv:1301.3656; Acta Math. Sin. (Engl. Ser.) 29 (2013), no. 11, 2037–2046.

[2] R. S. Phillips, *On linear transformations*, Trans. Amer. Math. Soc. 48 (1940), 516–541.

[3] M. Zippin, *The separable extension problem*, Israel J. Math. 26 (1977), 372–387.